

betaNB: Internal Tests

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Tests

```
#> test-betaNB-beta-nb-est
#> Call:
#> BetaNB(object = nb)
#>
#> Standardized regression slopes
#> type = "pc"
#>      est      se R  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> NARTIC  0.4951 0.0630 5 0.4361 0.4374 0.4431 0.5983 0.6014 0.6021
#> PCTGRT  0.3915 0.0587 5 0.2987 0.2993 0.3022 0.4318 0.4326 0.4328
#> PCTSUPP 0.2632 0.0870 5 0.1352 0.1372 0.1462 0.3434 0.3439 0.3441
#> Call:
#> BetaNB(object = nb)
#>
#> Standardized regression slopes
#> type = "pc"
#>      est      se R  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> NARTIC  0.4951 0.0630 5 0.4361 0.4374 0.4431 0.5983 0.6014 0.6021
#> PCTGRT  0.3915 0.0587 5 0.2987 0.2993 0.3022 0.4318 0.4326 0.4328
#> PCTSUPP 0.2632 0.0870 5 0.1352 0.1372 0.1462 0.3434 0.3439 0.3441
#> Call:
#> BetaNB(object = nb)
#>
#> Standardized regression slopes
#> type = "bc"
#>      est      se R  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> NARTIC  0.4951 0.0630 5 0.4359 0.4359 0.4360 0.5338 0.5733 0.5938
#> PCTGRT  0.3915 0.0587 5 0.2990 0.3014 0.3092 0.4326 0.4328 0.4329
#> PCTSUPP 0.2632 0.0870 5 0.1350 0.1354 0.1380 0.3420 0.3435 0.3440
#> Call:
#> BetaNB(object = nb)
#>
#> Standardized regression slopes
#> type = "bc"
#>      est      se R  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
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#> NARTIC 0.4951 0.0630 5 0.4359 0.4359 0.4360 0.5338 0.5733 0.5938
#> PCTGRT 0.3915 0.0587 5 0.2990 0.3014 0.3092 0.4326 0.4328 0.4329
#> PCTSUPP 0.2632 0.0870 5 0.1350 0.1354 0.1380 0.3420 0.3435 0.3440
#> Call:
#> BetaNB(object = nb)
#>
#> Standardized regression slopes
#> type = "bca"
#>      est      se R 0.05% 0.5% 2.5% 97.5% 99.5% 99.95%
#> NARTIC 0.4951 0.0630 5 0.4359 0.4359 0.4360 0.5343 0.5739 0.5943
#> PCTGRT 0.3915 0.0587 5 0.2992 0.3021 0.3104 0.4326 0.4328 0.4329
#> PCTSUPP 0.2632 0.0870 5 0.1349 0.1352 0.1371 0.3417 0.3433 0.3439
#> Call:
#> BetaNB(object = nb)
#>
#> Standardized regression slopes
#> type = "bca"
#>      est      se R 0.05% 0.5% 2.5% 97.5% 99.5% 99.95%
#> NARTIC 0.4951 0.0630 5 0.4359 0.4359 0.4360 0.5343 0.5739 0.5943
#> PCTGRT 0.3915 0.0587 5 0.2992 0.3021 0.3104 0.4326 0.4328 0.4329
#> PCTSUPP 0.2632 0.0870 5 0.1349 0.1352 0.1371 0.3417 0.3433 0.3439
#> Test passed with 1 success.
#> Call:
#> BetaNB(object = nb)
#>
#> Standardized regression slopes
#> type = "pc"
#>      est      se R 0.05% 0.5% 2.5% 97.5% 99.5% 99.95%
#> NARTIC 0.7622 0.0328 5 0.7353 0.7353 0.7355 0.8107 0.8137 0.8143
#> Call:
#> BetaNB(object = nb)
#>
#> Standardized regression slopes
#> type = "pc"
#>      est      se R 0.05% 0.5% 2.5% 97.5% 99.5% 99.95%
#> NARTIC 0.7622 0.0328 5 0.7353 0.7353 0.7355 0.8107 0.8137 0.8143
#> Call:
#> BetaNB(object = nb)
#>
#> Standardized regression slopes
#> type = "bc"
#>      est      se R 0.05% 0.5% 2.5% 97.5% 99.5% 99.95%
#> NARTIC 0.7622 0.0328 5 0.7353 0.7353 0.7353 0.8037 0.8116 0.814
#> Call:
#> BetaNB(object = nb)
#>

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#> Standardized regression slopes
#> type = "bc"
#>           est      se R  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> NARTIC 0.7622 0.0328 5 0.7353 0.7353 0.7353 0.8037 0.8116 0.814
#> Call:
#> BetaNB(object = nb)
#>
#> Standardized regression slopes
#> type = "bca"
#>           est      se R  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> NARTIC 0.7622 0.0328 5 0.7353 0.7353 0.7353 0.8016 0.8102 0.8135
#> Call:
#> BetaNB(object = nb)
#>
#> Standardized regression slopes
#> type = "bca"
#>           est      se R  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> NARTIC 0.7622 0.0328 5 0.7353 0.7353 0.7353 0.8016 0.8102 0.8135
#> Test passed with 1 success.

#> test-betaNB-delta-r-sq-nb-est

#> Call:
#> DeltaRSqNB(object = nb)
#>
#> Improvement in R-squared
#> type = "pc"
#>           est      se R  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> NARTIC 0.1859 0.0461 5 0.1327 0.1330 0.1344 0.2403 0.2425 0.2430
#> PCTGRT 0.1177 0.0521 5 0.0376 0.0383 0.0413 0.1576 0.1582 0.1583
#> PCTSUPP 0.0569 0.0358 5 0.0151 0.0158 0.0186 0.0985 0.0986 0.0986
#> Test passed with 1 success.
#> Test passed with 1 success.

#> test-betaNB-diff-beta-nb-est

#> Call:
#> DiffBetaNB(object = nb)
#>
#> Differences of standardized regression slopes
#> type = "pc"
#>           est      se R  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> NARTIC-PCTGRT 0.1037 0.1137 5 0.0033 0.0047 0.0113 0.2645 0.2647 0.2648
#> NARTIC-PCTSUPP 0.2319 0.1419 5 0.1248 0.1256 0.1289 0.4521 0.4642 0.4669
#> PCTGRT-PCTSUPP 0.1282 0.0803 5 -0.0090 -0.0079 -0.0031 0.1974 0.2042 0.2057
#> Test passed with 1 success.
#> Test passed with 1 success.

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#> test-betaNB-nb
#> Call:
#> NB(object = object, R = 6)
#>
#> The first six bootstrap covariance matrices.
#>
#> [[1]]
#>      [,1]      [,2]      [,3]
#> [1,] 1.0099124 0.53370896 0.47994262
#> [2,] 0.5337090 1.07034018 0.04362108
#> [3,] 0.4799426 0.04362108 0.99932510
#>
#> [[2]]
#>      [,1]      [,2]      [,3]
#> [1,] 0.9528211 0.44685052 0.48008660
#> [2,] 0.4468505 0.98824535 -0.02845785
#> [3,] 0.4800866 -0.02845785 1.04559557
#>
#> [[3]]
#>      [,1]      [,2]      [,3]
#> [1,] 1.0588165 0.540911948 0.480732629
#> [2,] 0.5409119 1.038247062 0.009921316
#> [3,] 0.4807326 0.009921316 1.002934084
#>
#> [[4]]
#>      [,1]      [,2]      [,3]
#> [1,] 1.0611349 0.48205132 0.50436155
#> [2,] 0.4820513 0.99659705 0.00559072
#> [3,] 0.5043616 0.00559072 0.96031537
#>
#> [[5]]
#>      [,1]      [,2]      [,3]
#> [1,] 1.1187257 0.55970856 0.56140845
#> [2,] 0.5597086 1.02203524 0.02477157
#> [3,] 0.5614085 0.02477157 1.12240179
#>
#> [[6]]
#>      [,1]      [,2]      [,3]
#> [1,] 1.0318549 0.54098946 0.47090224
#> [2,] 0.5409895 1.13249519 0.03964982
#> [3,] 0.4709022 0.03964982 0.96126856
#>
#> Test passed with 1 success.
#> test-betaNB-p-cor-nb-est
#> Call:

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#> PCorNB(object = nb)
#>
#> Squared partial correlations
#> type = "pc"
#>      est      se R  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> NARTIC  0.4874 0.0872 5 0.3843 0.3863 0.3954 0.6010 0.6038 0.6044
#> PCTGRT  0.3757 0.1251 5 0.1988 0.2005 0.2079 0.5176 0.5267 0.5288
#> PCTSUPP 0.2254 0.1330 5 0.0762 0.0793 0.0932 0.4088 0.4110 0.4115
#> Test passed with 1 success.
#> Test passed with 1 success.

#> test-betaNB-r-sq-mc-est

#> Call:
#> RSqNB(object = nb)
#>
#> R-squared and adjusted R-squared
#> type = "pc"
#>      est      se R  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> rsq 0.8045 0.0290 5 0.7874 0.7879 0.7902 0.8580 0.8588 0.8590
#> adj 0.7906 0.0311 5 0.7722 0.7727 0.7752 0.8478 0.8487 0.8489
#> Test passed with 1 success.
#> Call:
#> RSqNB(object = nb)
#>
#> R-squared and adjusted R-squared
#> type = "pc"
#>      est      se R  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> rsq 0.5809 0.0507 5 0.5406 0.5407 0.5409 0.6574 0.6621 0.6631
#> adj 0.5714 0.0519 5 0.5302 0.5302 0.5305 0.6496 0.6544 0.6555
#> Test passed with 1 success.

#> test-betaNB-s-cor-nb-est

#> Call:
#> SCorNB(object = nb)
#>
#> Semipartial correlations
#> type = "pc"
#>      est      se R  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> NARTIC  0.4312 0.0540 5 0.3642 0.3647 0.3665 0.4901 0.4924 0.4929
#> PCTGRT  0.3430 0.0856 5 0.1939 0.1953 0.2019 0.3970 0.3977 0.3979
#> PCTSUPP 0.2385 0.0809 5 0.1230 0.1248 0.1329 0.3139 0.3140 0.3140
#> Test passed with 1 success.
#> Test passed with 1 success.
#> [[1]]
#> [[1]][[1]]

```

```

#> [[1]][[1]]$value
#> [[1]][[1]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[1]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[2]]
#> [[1]][[2]]$value
#> [[1]][[2]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[2]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[3]]
#> [[1]][[3]]$value
#> [[1]][[3]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[3]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[4]]
#> [[1]][[4]]$value
#> [[1]][[4]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[4]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[5]]
#> [[1]][[5]]$value
#> [[1]][[5]]$value[[1]]
#> [1] TRUE
#>
#>
#>

```

```
#> [[1]][[5]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[6]]
#> [[1]][[6]]$value
#> [[1]][[6]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[6]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[7]]
#> [[1]][[7]]$value
#> [[1]][[7]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[7]]$visible
#> [1] TRUE
```

Environment

```
ls()
```

```
#> [1] "nas1982" "root"
```


Class

```
#> [[1]]  
#> [1] "data.frame"  
#>  
#> [[2]]  
#> [1] "root_criterion"
```

References

R Core Team. (2026). *R: A language and environment for statistical computing*. R Foundation for Statistical Computing. Vienna, Austria. <https://www.R-project.org/>